Name:		Index Number:		Class:	

DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H2 BIOLOGY — 9744/03

Paper 3 Long Structured and Free Response Questions

20 September 2018

2 hours

Additional Materials: Writing paper

INSTRUCTIONS TO CANDIDATES:

DO NOT TURN THIS PAGE OVER UNTIL YOU ARE TOLD TO DO SO.

READ THESE NOTES CAREFULLY.

Section A Long Structured Questions

Answer **all** questions.

Write your answers on space provided in the Question Paper.

Section B Free-Response Questions

Answer **one** question. Your answer to Section B must be in continuous prose, where appropriate. Write your answers on the writing paper provided. **Answer each part (a) and (b) on a fresh piece of writing paper.**

INFORMATION FOR CANDIDATES

Essential working must be shown.

The intended marks for questions or parts of questions are given in brackets [ ].

For Examiner's Use	
Section A	
1	/ 27
2	/ 8
3	/ 15
Section B	/ 25
Total	/ 75
Grand Total [260]	

For Examiner's use

Section A: Long Structured Questions (50 marks)

Answer **all** questions in this section.

Question 1

Fig. 1.1 shows the life cycle of Dengue virus.

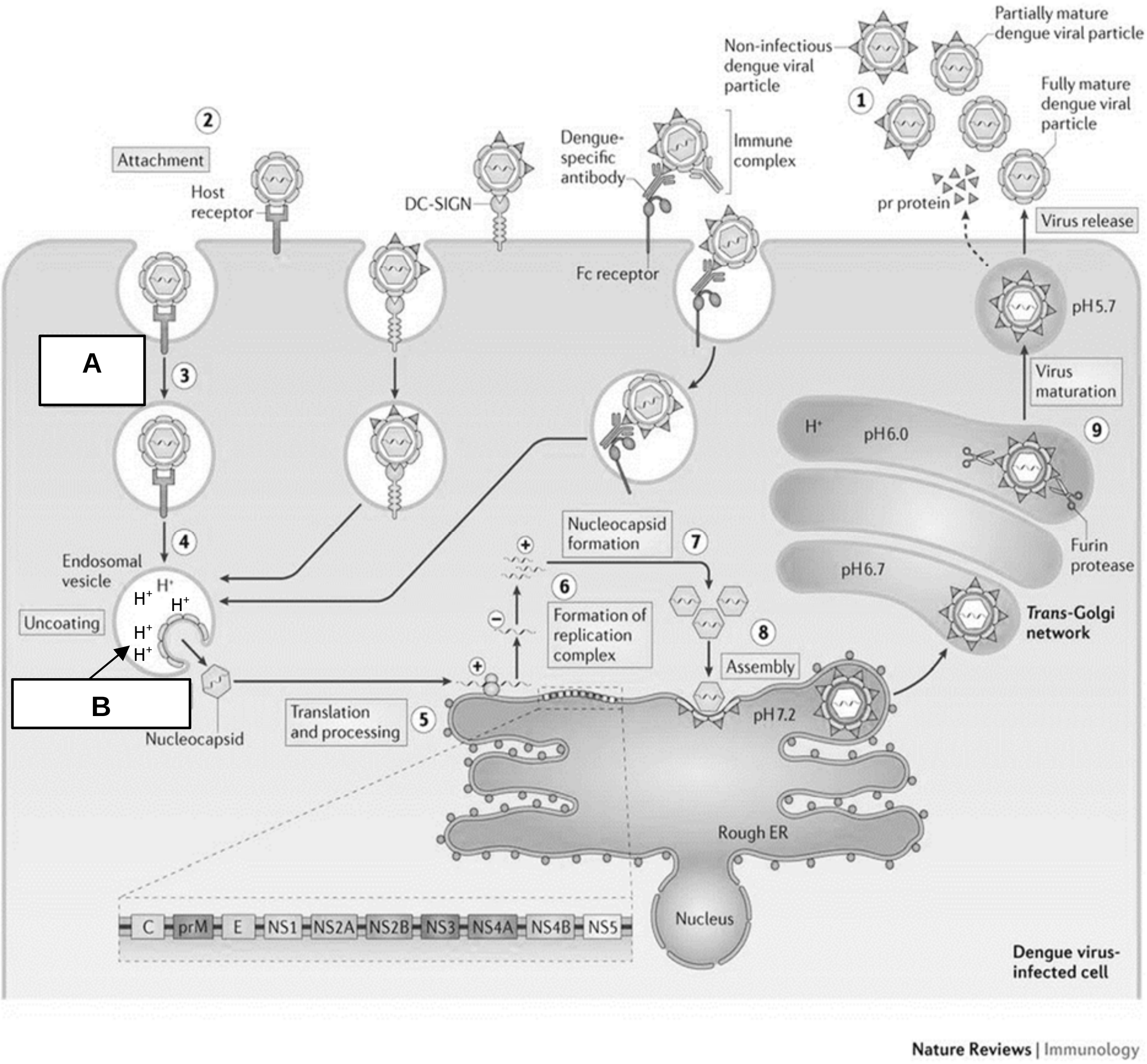

Fig. 1.1

(a) **(i)** Identify processes A and B. [2]

Process A: ______________________________

Process B: ______________________________

For Examiner's use

(ii) With reference to Fig. 1.1, identify and explain the type of genome found in Dengue virus. [2]

(iii) Describe the main events from step 8 to step 9. [4]

(b) A dengue virus vaccine, Dengvaxia, was recently developed not only to prevent first infections, but also to avoid severe disease that may occur upon second and third infections. This consequence, called antibody dependent enhancement (ADE), now appears to be caused by the vaccine itself.

Primary dengue virus infection may lead to a disease that includes fever, headache, muscle, joint pain and skin rash. In about 1 in 10,000 infections, severe disease can develop which involves severe bleeding, shock, and hemorrhagic fever. In a secondary infection involving a different serotype, the risk of severe disease is much higher, approaching 1 in 100.

After a 3-phase clinical trial, it is now recognize that serious dengue, defined as requiring hospitalization, has occurred in two populations: vaccinated **seronegative** children, and monotypic dengue virus immune children who received a placebo. In the seronegative children, the vaccine acted like a primary dengue virus infection, causing severe disease after an infection acquired 'in the wild'. In immune children who received the placebo, a second acquired infection led to ADE.

During year 5 after the third dose of vaccine, 295 of 20,439 children were hospitalized with dengue virus infection. Severe dengue was also observed during years 2-4 after the third dose.

(i) Using information from paragraph 3, explain the term "**seronegative** children". [2]

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(ii) Using information from paragraph 4, suggest why many countries are not enforcing the use of dengue vaccine. [2]

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For Examiner's use

(iii) Nearly all children who received Dengvaxia produced neutralizing antibodies against all four serotype of dengue virus. With reference to Fig. 1.2, explain why are they still affected by serious disease. [5]

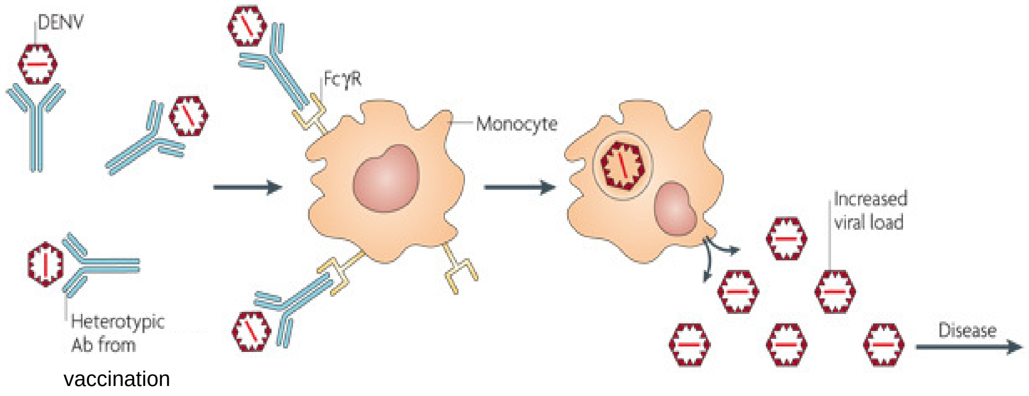

Fig. 1.2

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For Examiner's use

During an overseas trip to the Sahara Desert, Howard was accidently stung by one of the most venomous scorpions in the world, the Deathstalker. The venom of the Deathstalker scorpion contains a potent toxin known as chlorotoxin. Fig. 1.3 shows the effect of the chlorotoxin from the Deathstalker scorpion.

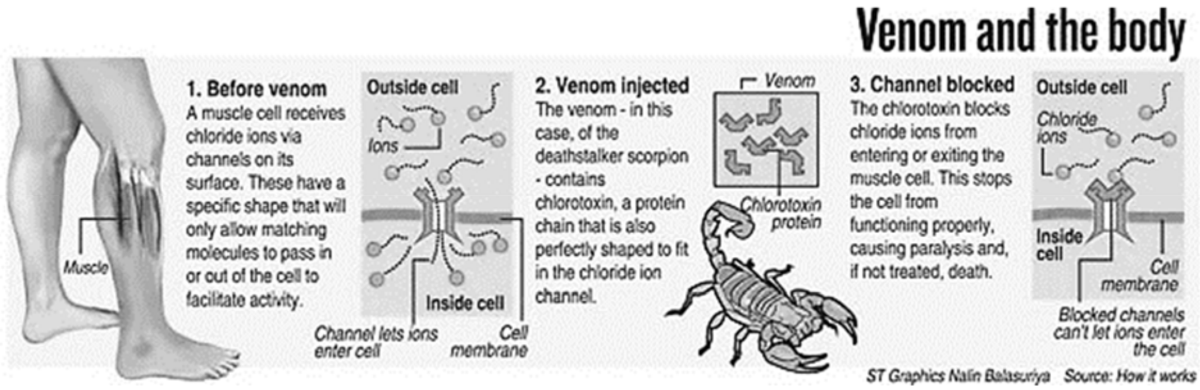

Fig. 1.3

In an attempt to save Howard's life, doctors from a nearby hospital gave Howard an injection of antivenom containing antibodies that bind specifically to chlorotoxin. The effect of the antivenom is shown in Fig. 1.4.

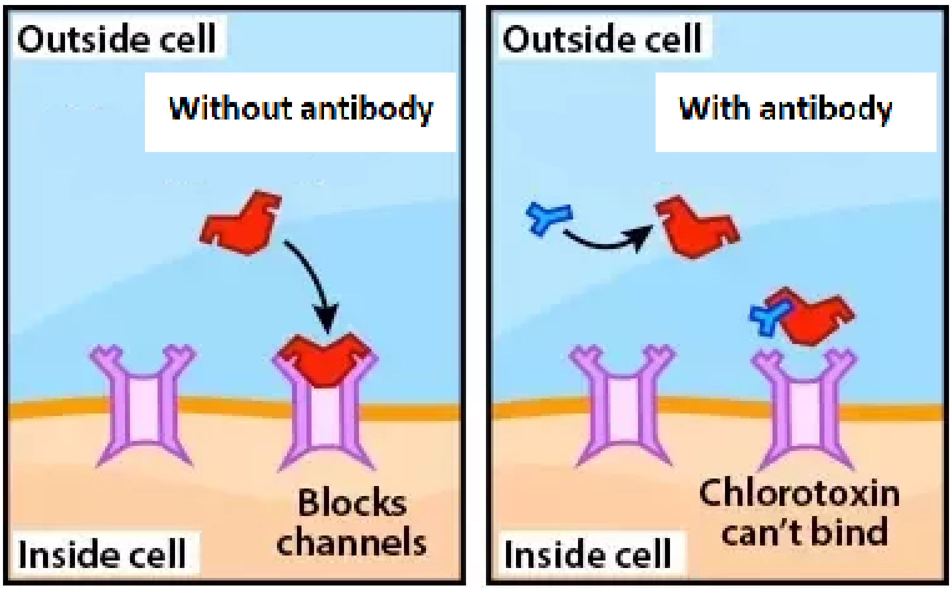

Fig. 1.4

(c) **(i)** State the type of immunity conferred by the antivenom. [1]

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For Examiner's use

(ii) Using the information provided in Fig. 1.3 and Fig. 1.4, explain how the antivenom helped save Howard's life. [4]

(d) After his near-death experience, Howard had an epiphany and decided to pursue his lifelong ambition to be a professional snake charmer. Using his life savings, he purchased a particularly venomous King cobra for his future performances from a local snake dealer.

Before Howard can embark on his new career, he will need to determine how to deal with accidental envenomation (the process by which venom is injected by the bite or sting of a venomous animal). One way is to ensure that he keeps a vial of antivenom (containing antibodies that bind to the toxins present in the King cobra's venom) with him at every performance. However, according to the local snake dealer, several snake handlers have managed to build up an immunity through the practice of mithridatism by regularly injecting themselves with non-lethal doses of the venom.

Discuss the use of mithridatism in place of antivenom to counteract the toxins in the King cobra's venom. [5]

Total: [27]

Question 2

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Coral reefs are affected by climate change. Fig. 2.1 shows how sea water temperatures have varied over the last century or so at the Great Barrier Reef. 0˚C is considered the average sea water temperature. Rising sea level temperatures have been suggested as a reason for more frequent coral bleaching events. Fig. 2.1 shows the trend for coral bleaching events globally since 1980.

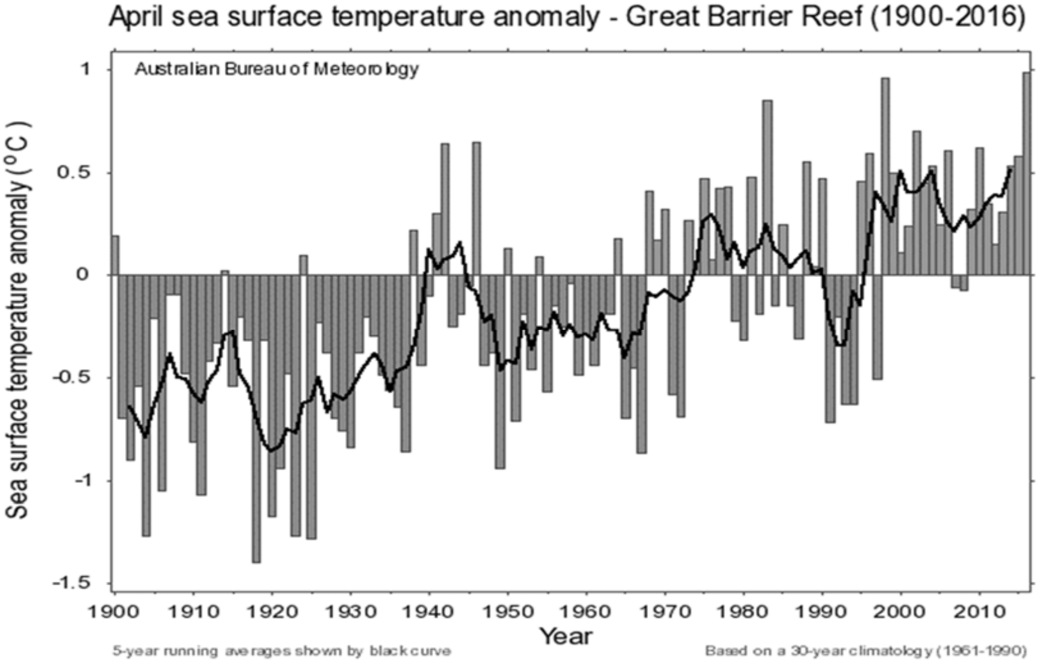

Fig. 2.1

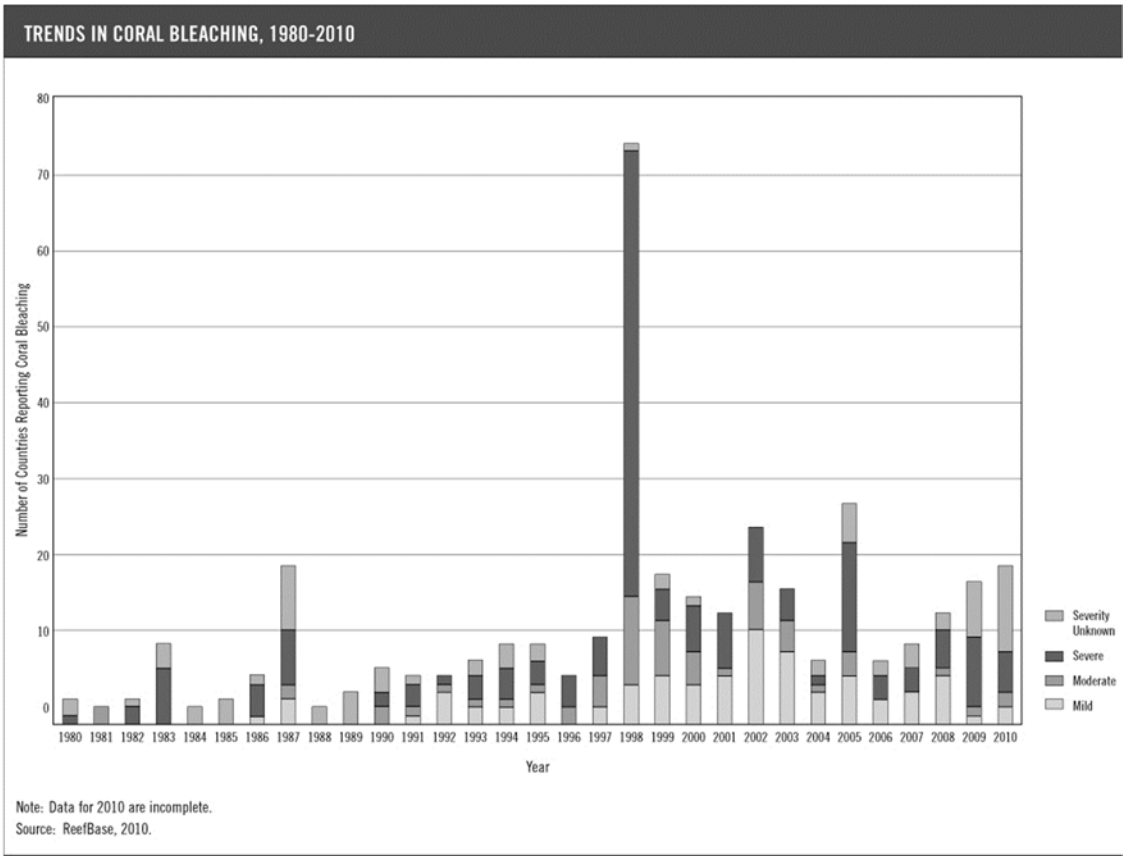

Fig. 2.2

For Examiner's use

(a) With reference to Fig. 2.1 and 2.2, explain whether the data provides sufficient evidence that the more frequent bleaching events seen in coral reef ecosystems over the last 20 years are due to rising water temperatures. [6]

(b) Other than rising water temperatures, suggest two other ways climate change has impacted coral reefs. [2]

Total: [8]

For Examiner's use

Question 3

(a) Fig. 3.1 shows the peptidoglycan cell wall of the prokaryote *Escherichia coli*.

The carbohydrate backbones are made up of two different kinds of sugar: N-acetylmuramate (NAM) and N-acetylglucosamine (NAG). NAM and NAG alternate along the chain and differ from glucose only at the C2 and C3 position.

Fig. 3.2 shows the molecular structure of NAM and NAG.

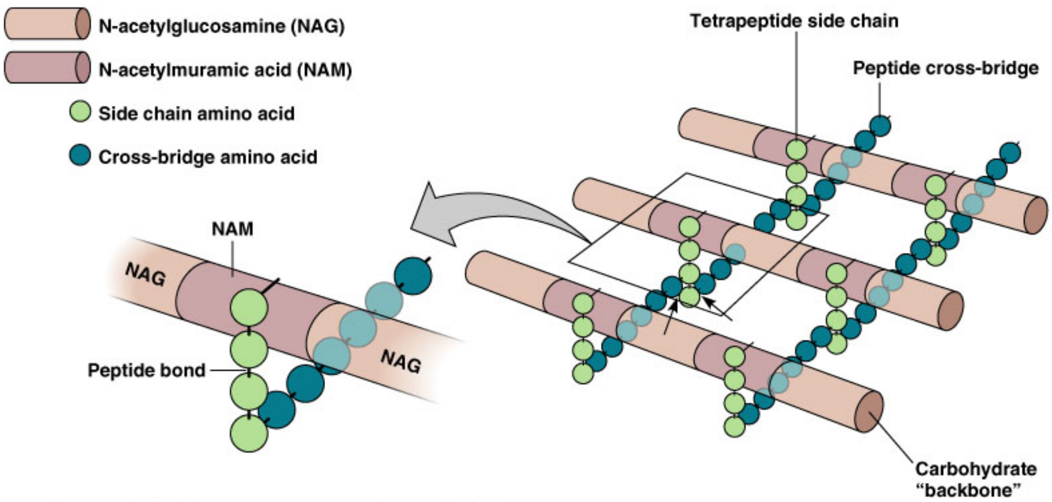

Fig. 3.1

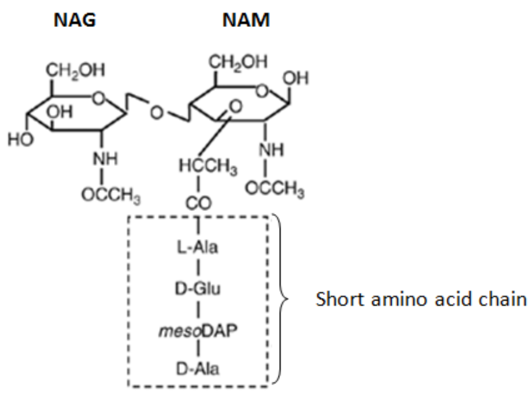

Fig. 3.2

With reference to Fig 3.1 and Fig. 3.2, describe two ways in which the structure of peptidoglycan:

(i) is similar to cellulose. [2]

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(ii) differs from cellulose. [2]

Features	Peptidoglycan	Cellulose

(iii) The enzyme lysozyme is found in many pharmaceutical products like eye drops and throat lozenges that treat bacterial infection. Lysozyme is a glycoside hydrolase.

Suggest how the enzyme could be used to treat bacterial infection. [2]

For Examiner's use

(b) The oil stored in the seeds of the castor oil plant *Ricinus communis* has been used in the production of pharmaceutical products such as zinc and castor oil cream, and as a lubricant. However, *Ricinus* seeds also produce a poisonous protein ricin, which is classified as an agent of bioterrorism.

Fig. 3.3 shows the structure of ricin which consists of two polypeptides – the A chain of ricin (RTA, ricin toxin A) joined to a B chain (RTB, ricin toxin B).

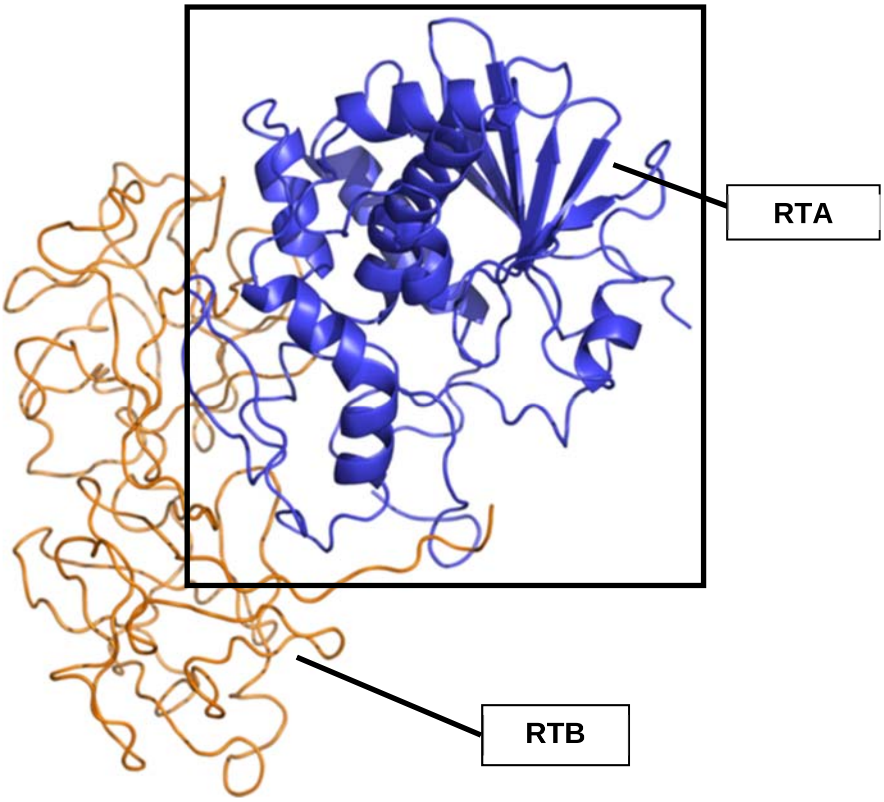

Fig. 3.3

(i) The RTB component of ricin has two binding sites specific for galactose. This enables RTB to bind to cell surface components that contain galactose. This is the first step of cellular intoxication.

Identify a structure found on the mammalian cell surface that makes it a good target for ricin. [1]

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The RTA component of ricin is an enzyme that damages an essential component of the protein synthesis machinery – the ribosome. RTA catalyses the removal of one particular nitrogenous base (adenine) in a crucially important section of ribosomal RNA of the large subunit of eukaryotic ribosomes. This results in eventual cell death.

The two polypeptides – RTA and RTB are covalently linked by a disulfide bond between two cysteine residues. Reduction of ricin into its subunits, RTA and RTB, is catalysed by protein disulfide isomerase (PDI), an enzyme that resides in the lumen of the endoplasmic reticulum. PDI breaks the interchain disulfide bond.

Fig. 3.4 illustrates the reduction process catalysed by PDI.

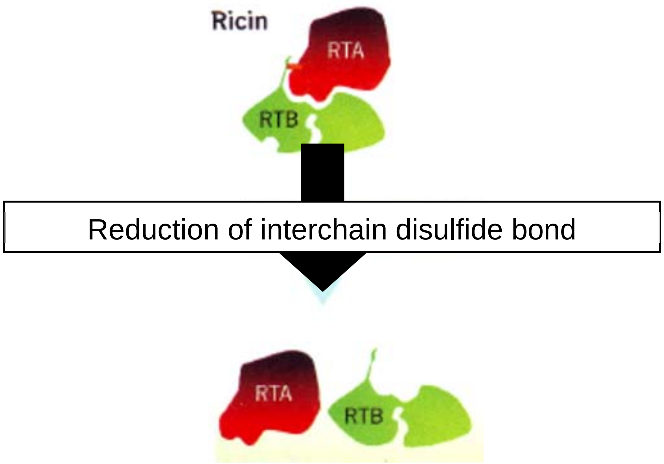

Fig. 3.4

(ii) Explain why the reduction process shown in Fig. 3.4 is necessary before RTA can exhibit its catalytic activity. [2]

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(iii) Suggest one way in which plant ribosomes in the seed cells are protected from the effect of ricin. [1]

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(c) Enzyme **X** digests a sugar and the reaction was carried out in a conical flask. Enzyme **X** may be immobilised in alginate beads.

Fig. 3.5 shows a comparison between the activity of free enzyme **X** and immobilised enzyme **X** over a range of temperatures. Equal concentrations of free enzyme **X** and immobilised enzyme **X** were used.

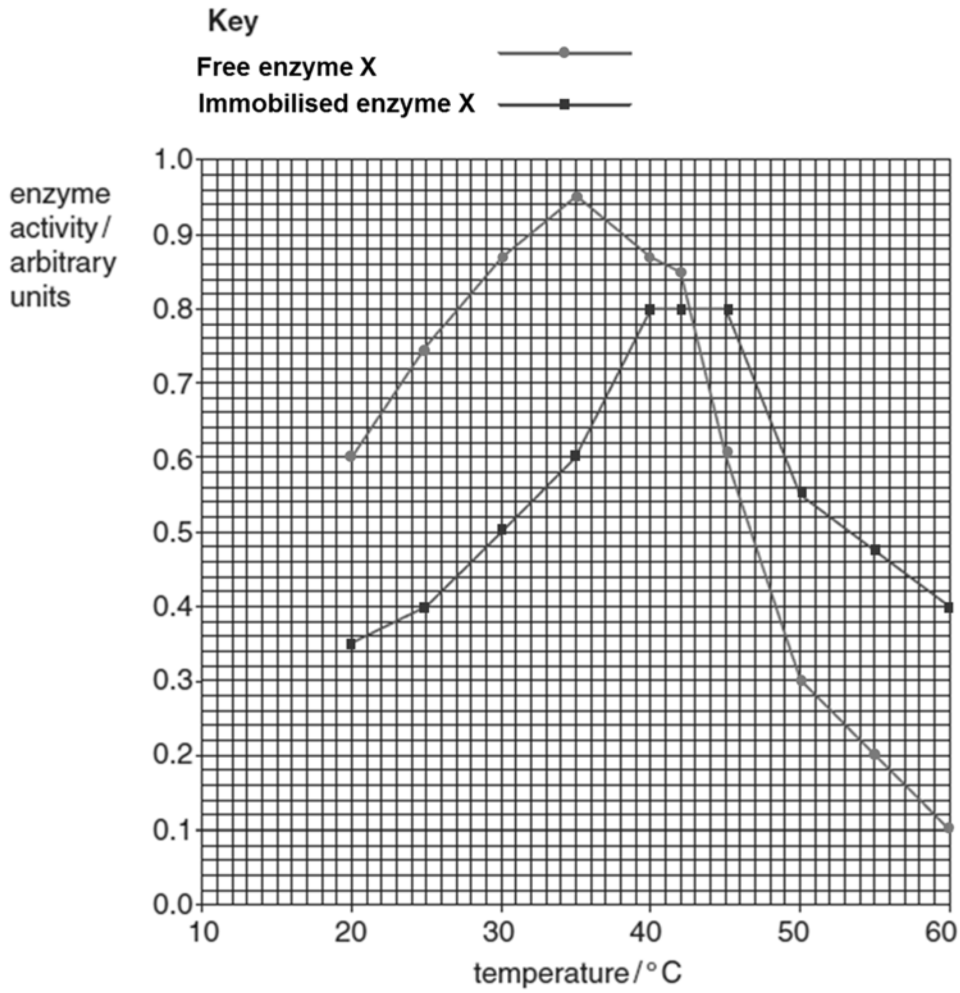

Fig. 3.5

(i) Explain the effect of temperature on the activity of free enzyme **X**. [3]

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For Examiner's use

(ii) Explain the differences between the activity of free enzyme **X** and immobilised enzyme **X** up to 40 °C. [2]

Total: [15]

Section B: Free-Response Question (25 marks)

Answer only **one** question.
Write your answers on the writing paper provided.
Answer each part (a) and (b) on a fresh piece of writing paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.
A **NIL RETURN** is required.

Question 4

(a) Explain how the various aspects of DNA replication mechanism are important in the design of PCR technique and what additional considerations are needed for a PCR process. [10]

(b) Discuss the ethical implications of the application of stem cells in research and medical applications and how human induced pluripotent stem cells (iPSCs) overcome some of these issues. [15]

Total: [25]

OR

Question 5

(a) Describe the mutation that give rise to sickle cell anaemia. [10]

(b) *Staphylococcus aureus* is a common bacterium, found on human skin. If it enters the body through wounds or surgery, it can cause serious infection.

There are many strains of *Staphylococcus aureus*. Most of these strains were sensitive to penicillin when the antibiotic was first introduced in the 1940s. However, by the late 1950s, over 90% of *Staphylococcus aureus* strains were resistant to penicillin. The antibiotic methicillin was then used to treat infection by *Staphylococcus aureus*. Now, there are at least 15 different strains of Methicillin Resistant *Staphylococcus aureus* (MRSA).

Explain how 15 different strains of MRSA may arise from a single strain of *Staphylococcus aureus*. [15]

Total: [25]

END OF PAPER